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Magazine

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REVIEWING THE LATEST IN IPTV PRODUCTS, TECHNOLOGIES AND APPLICATIONS

Internet TV Set Top Boxes

01-10
Jan 2010
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**IPTV Global Forecast
Market Update**

PLUS

User Controlled TV Advertising
Internet TV Set Top Boxes
IPTV Subscription Rate
Plans Update

**IPTV Channel Hosting Companies
Buyers Guide
Inside!**

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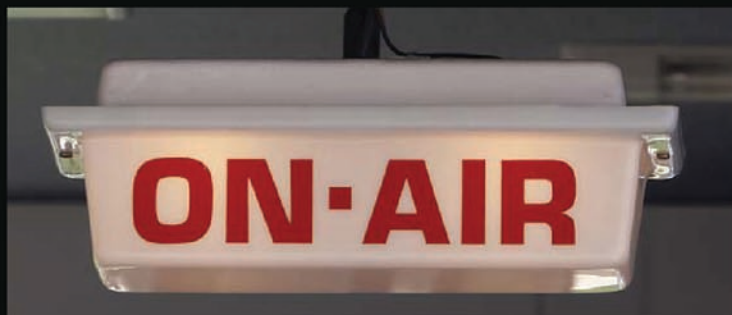
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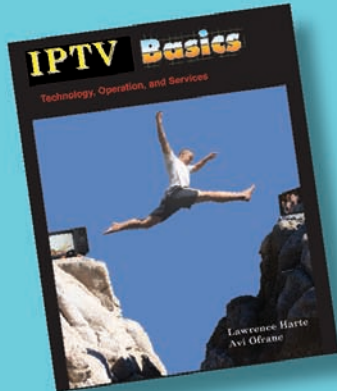
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IP Television Basics

IP Television Basics: Technology, Operation, Economics, and Services provides an understanding of how IP television technology operates, what applications it can offer, the costs & benefits of packetized video systems, and the new services that can be provided by IP Television systems.

Authors: Harte, Flood
384 pages, \$39.99

Introduction to IP Television



This book explains how and why people and companies are using IP television and Internet television services.

\$14.99

Introduction to Mobile Video



Described are the key mobile video applications including sending video clips, live TV, video messaging and multi party gaming.

\$19.99

Introduction to Premises Distribution Networks



This book covers the different types of premises distribution networks (PDNs) that distribute audio, data and video through a home or business.

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Editor's Viewpoint



TV advertising regulation defines how much advertising TV broadcasters can offer, as well as authorized types and suitable broadcast times. TV advertising regulation can be significantly more restrictive in some countries, such as the UK, which can reduce or even collapse the ability of TV broadcasters to offer broadcast services. Excessive regulation of the TV broadcasting industry can result in fewer programming choices, and even the loss (bankruptcy) of TV broadcasters.

TV broadcasters have been experiencing increases in content costs in conjunction with reductions (non-increases) in advertising revenues. In general, advertising budgets have been shifting towards Internet Marketing channels where there are less regulations regarding promotions.

As more content sources become available to the consumer through new channels such as mobile video and Internet TV, consumers will shift from traditional broadcasters to alternative sources. This reduction in viewers will result in decreased advertising revenues.

To help TV broadcasters survive, the regulatory requirements for TV advertising should be reduced, or similar regulations should be applied to Internet advertising. The best solution may be a combination of allowing more types of advertising on television systems and better control over promotional regulations on the Internet.

TV Advertising Regulation - Not Balanced with Internet TV Advertising

Lawrence Harte, Editor

IP Television Expert Writers

What makes a magazine successful is the value of its content. Our expert writers cover marketing, technology and business issues that are critical to the success of IP television systems and services.



Robert Belt
Marketing

Mr. Belt is a new product business development, marketing and sales expert for communication products. Robert specializes in assisting international and OEM companies in finding, qualifying, establishing meetings, defining product requirements, negotiation of sales contracts and follow-up customer communication. He has more than 20 years of experience in product definition, engineering specification, design and contract negotiation for non-standard and new products. Mr. Belt has defined and located for strategic partnering, initiated discussions for technology partnering and drafted alliance agreements. Robert's clients have included Alps, Motorola, Nokia, Wavetek, Norand, Trimble, Mitsubishi, Panasonic, Fujitsu, Uniden, NEC, Qualcomm, Novatel, JRC, Apple, Omnipoint, NYNEX, Bell Atlantic, SONY and hundreds of other companies.



Bud Bates
Technology

Regis (Bud) Bates is a wireless systems expert who specializes in network operations and planning for telecommunications and management information systems. As president of TC International Consulting, he performs Strategic Planning, Business Continuity Planning and Technology Innovation for his client companies. Mr. Bates has helped fortune 100-500 companies design, setup, and manage LANs and WANs using SONET, ATM, MPLS, and VPN architectures. He specializes in the setup of mobile communication systems and developing the processes necessary to ensure the reliable restoration of networks when failures occur. Bud is a sought after professional instructor and he teaches using both Instructor-led (ILT) and Virtual classroom learning (VCL) formats. Bud Bates authored over fifteen technology-oriented books, many of which were best sellers for McGraw-Hill. Bud received his degree in Business Management from Stonehill College (BS) in Easton, MA and completed an MBA in Finance at St. Joseph's University in Philadelphia (except the thesis).



Lawrence Harte
Business

Mr. Harte has over 29 years of technology analysis, development, implementation, and business management experience. Mr. Harte has worked for leading companies including Ericsson/General Electric, Audiovox/Toshiba and Westinghouse and has consulted for hundreds of other companies. Mr. Harte continually researches, analyzes, and tests new communication technologies, applications, and services. He has authored over 60 books on communications technologies and business systems covering topics such as IP television, mobile telephone systems, data communications, voice over data networks, broadband, prepaid services, billing systems, sales, and Internet marketing. Mr. Harte holds many degrees and certificates including an Executive MBA from Wake Forest University (1995) and a BSET from the University of the State of New York, (1990).



Roger McGarrahan
Content
Licensing

Roger McGarrahan is co-founder and General Manager of PathFinder World Video LLC which licenses linear channel and VOD programming from ethnic and niche television networks to CATV, Telco IPTV, Broadband IPTV, Mobile and Hospitality television service providers. Prior to that Roger was CEO of Thomson Broadcast & Multimedia, Inc. (Thomson/Grass Valley) in charge of North America operations and previously its General Counsel. Earlier Roger was legal counsel for COMSAT RSI which specialized in the design and delivery of satellite communication systems. In total, Roger has twenty years experience as executive management, operations management, and corporate counsel in the broadcast, satellite and telecommunications industries.



Michael Sommer
Consumer
Electronics

Michael H. Sommer - The "Gadget Guy" Technology Commentator is a consumer electronics industry expert. Mr. Sommer regularly appears on several television stations as the Gadget Guy and is a sought after technology evaluation and marketing expert. His words and industry findings are referenced in many leading industry publications including USA Today, N.Y. Times and Telecom Business magazine. Mr. Sommer has been on the communication staff of the Winter Olympics and he is a staff expert writer for IP Television Magazine. He has been a consultant for hundreds of consumer electronics product developers ranging from high-tech start-ups to fortune 100 multinational companies. His clients include Motorola, Cendant Corporation, Sony, and other leading edge companies. Mr. Sommer attended the University of Hartford majoring in communications and he specializes in working with executives from fortune 1000 companies providing them with an understanding of consumer electronics device requirements and marketing programs.



Eric Stasik
Patents &
Legal

Mr. Eric Stasik is the director of Patent08, an expert consulting firm located in Stockholm, Sweden providing patent engineering, business development, and licensing services to small and medium-sized enterprises. He is an expert in helping firms develop patent and licensing strategies that support their business objectives. He is the author of several books on patent strategy and maintains a well-respected blog (www.patent08.com) on the business aspects of developments on patent law and practice. Mr. Stasik is an engineer; he is not an attorney at law and does not provide legal advice.



Avi Ofrahe
Billing
Systems

Avi Ofrahe is the president and CEO, and a master instructor of The Billing College. Mr. Ofrahe founded The Billing College in 1996 to address the converging market trends associated with telecommunications Billing and Customer Care. Mr. Ofrahe began his career in 1977 as an analyst with the IBM Corporation, designing and implementing manufacturing systems. Throughout his extensive career, Mr. Ofrahe has been involved in all aspects of the industry, including strategic planning, RFP processing, vendor evaluation and selection, business process engineering, business/systems analyses, project management, implementation, operations, quality assurance, and executive management. Since 1982, Mr. Ofrahe has concentrated exclusively on the telecommunications industry, in which he is now a recognized expert and master instructor in Billing and Customer Care. Mr. Ofrahe lectures extensively in the US and in Europe on Billing and Customer Care issues, strategies, methodologies, and practices and he is a frequent speaker at major industry conferences. He has authored several leading books on billing systems. Mr. Ofrahe holds a BS, Computer Science, from Pennsylvania State University.

Market Update

Global IPTV Predictions 2009-2013

By: MRG

According to Multimedia Research Group (MRG), in 2008 the actual IPTV subscribers ended up at about 1 million over its last forecast in late 2008, or 21.3 million, resulting in projected subscriber growth of 26.9 million in 2009 to over 81 million in 2013. Combined (last-mile) CapEx revenue plus service revenue will grow from US\$9.7 billion in 2009 to US\$25.6 billion in 2013.

As the IPTV market matures, many innovations are emerging, including Service Providers' turning to Over-the-Top Video applications to supplement their video-on-demand offerings. Technical upgrades also contribute to growth, including DVRs, High-definition programming, MPEG-4/H.264, and first class system integration.

685 companies worldwide were identified as deploying IPTV services. Although the number of Service Providers have increased since the last report, some have also been removed because they either stopped providing IPTV services, were duplicates or they've merged with another company.

The biggest growth region came from the ROW category, which went up from 64 companies to 84. Countries like Colombia, Qatar, United Arab Emirates, Montenegro and the Russian federation have seen new growth in their operations, and, the ROW region will be among the fastest-growing market from 2009 to 2013 with a 29% CAGR.

Figure 1 shows the global IPTV subscriber forecast for 2009-2013. MRG is forecasting that the number of global IPTV subscribers will grow from 26.7 million in 2009 to 81 million in 2013, a compound annual growth rate of 32%. The five-year forecast has gone down from the Fall 2008 Global Forecast where MRG predicted 89.1 million subscribers by 2012.

One indicator that new subscriptions will remain strong is the Q1/2009 IPTV subscriber growth of 583,000 combined for U.S. Verizon and AT&T compared with 114k new subs added by the two largest U.S. Cable Operators, Comcast and Time Warner for the same

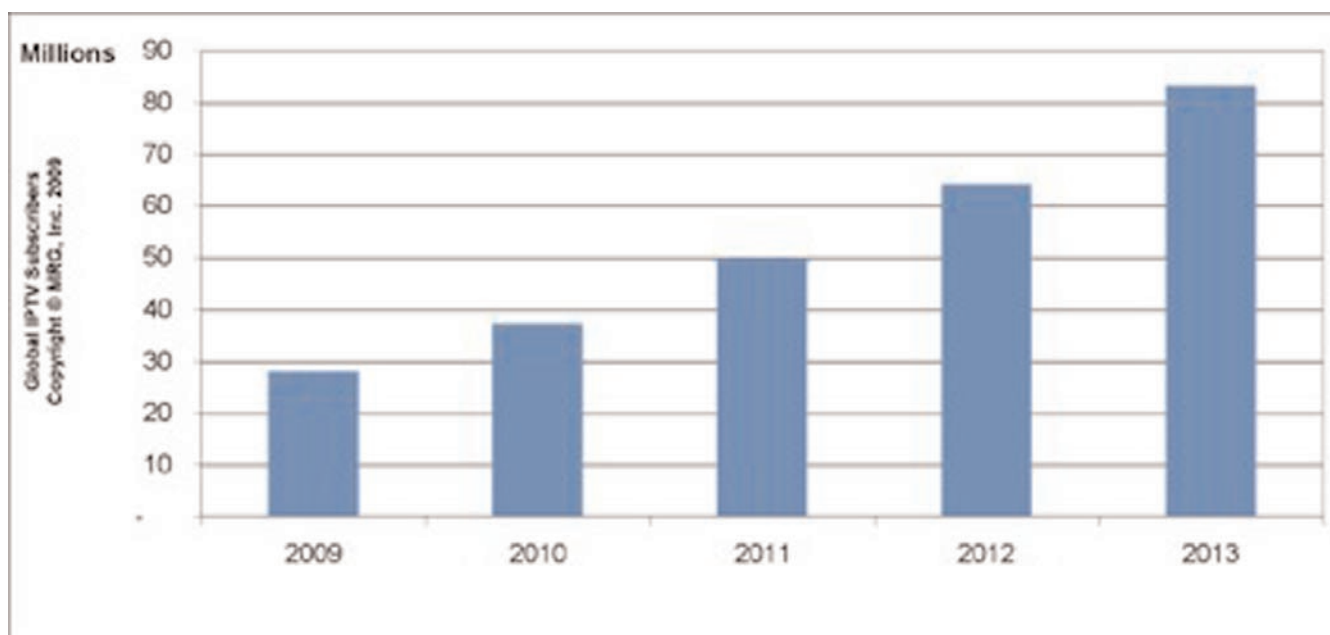


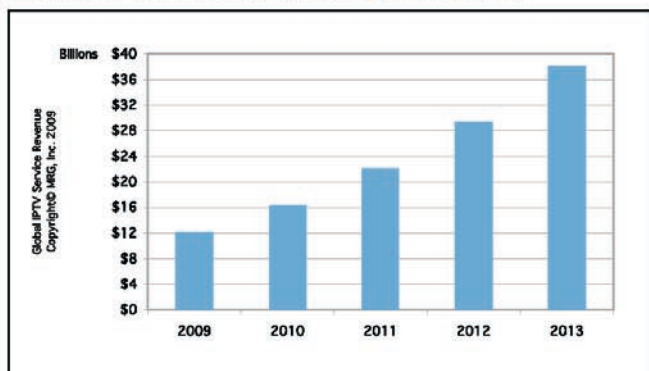
Figure 1: Global IPTV Subscriber Forecast



IPTV Global Forecast 2009 to 2013

November 2009

Global IPTV Service Revenue Forecast



Source: Copyright © 2009 MRG, Inc.

MRG's new bi-annual IPTV Global Forecast Report — 2009 to 2013 for November 2009 displays much more information in strategies and new service mixes being added to IPTV offerings.

This report is designed for senior management at hardware/software suppliers, Telcos, IPTV Operators, venture capital firms and financial institutions.

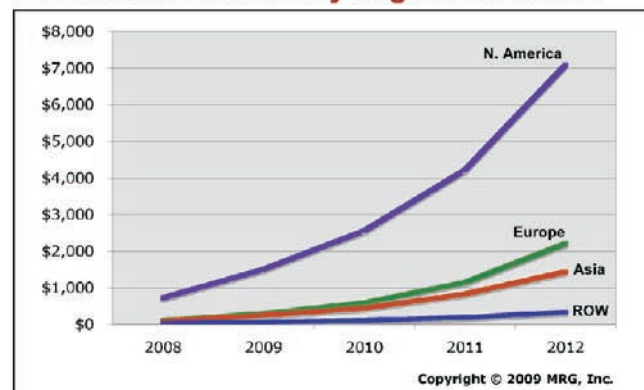
Tracking over 800 total IPTV Operators worldwide, the report analyzes service revenue (chart above), capital spending and subscriber growth for Asia, Europe, North America, and the Rest of the World.

It also breaks down CapEx by seven IPTV product sectors, including Access Systems, Video Headends, Video-on-Demand, Set-top Boxes, Middleware, Content Protection/Digital Rights Management, and System Integration/Professional Services. The report also includes IPTV capital spending detail of the top 25 global service providers.

Global OTT Video Services & Forecast—2008-2012: Best Practice, ROI, & ISTB Analysis

August 2009

OTT Service Revenue By Region: 2008-2012



Source: Copyright © 2009 MRG, Inc.

As TV-centric video content on the Internet (or OTT/Over-the-Top Video) grows in popularity and also increases in quality, the burning questions become:

- How can Service Providers (SPs) benefit from this?
- How and why will these services play out in North America, Europe, Asia and Rest-of-World Markets?
- What business models work best for the SP in both the short-term and long-term, and why?
- What platform works best for consumers in North America, Europe and Asia and why (including a ROI Analysis)?
- How does the OTT service differ from the hybrid service model described in MRG's earlier report?

This report answers these questions and or provides the tools necessary to answer them. These answers are especially important in an uncertain economy because Service Providers must find new services that can be deployed quickly, with low cost, and with some predictability about ROI.

To Order

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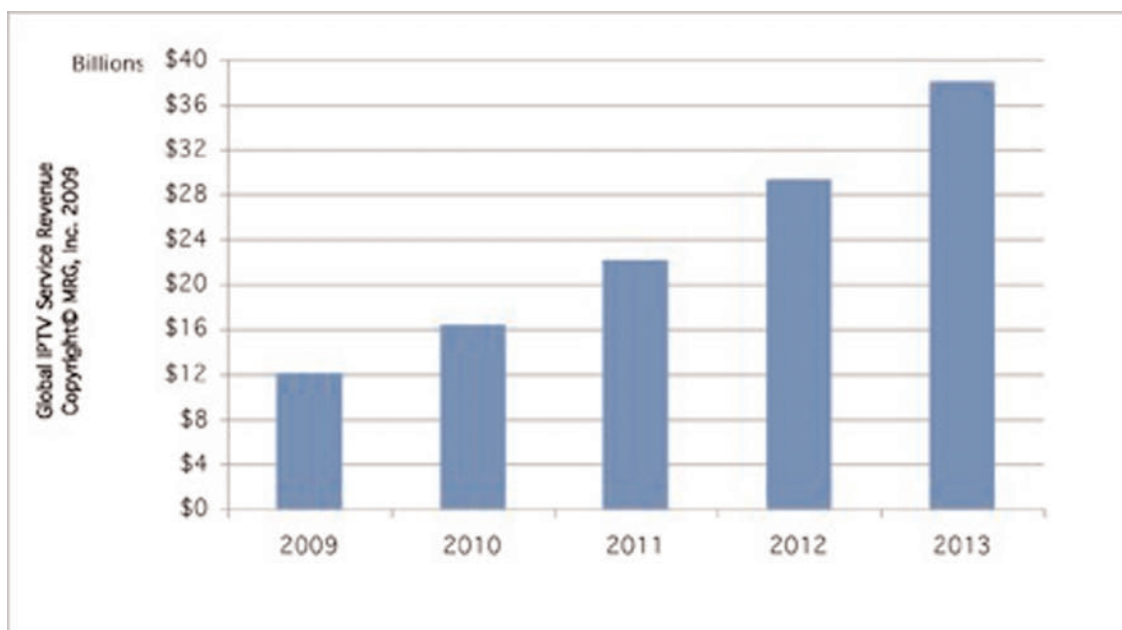


Figure 2 : Global IPTV Revenue Forecast

period. Also a signal of new growth is the number of new IPTV Operators in Eastern Europe and the Rest-of-World (ROW) region.

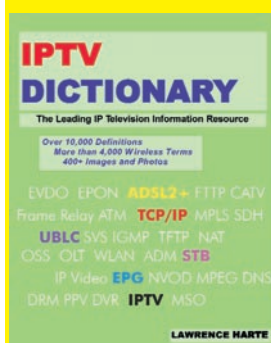
Previously both Europe and Asia have battled for the region with the largest number of IPTV subscribers in 2013. However in this report, we've again forecasted that Europe will remain the largest region, with almost 35 million subscribers in 2013, or about 43% of the total worldwide IPTV subscriber total. Asia is next with almost 27 million subscribers in 2013, or 34% of the worldwide total.

Forecasts for Asia had to be lowered because of slower growth rates in Korea and Japan, although the total growth rate in our forecast is still a healthy 34%. China will take the lead in Asia, with 45% of that region's total and 15% of the worldwide IPTV subscriber total. North America will have the top two largest IPTV Service Providers globally by 2013, however 21% of the overall global IPTV market. The Rest of the World region will see aggressive growth from 2009 to 2013 (29.3% growth rate), but it is starting from a very small base, and will represent only 3% of the IPTV subscriber base in 2013.

In terms of service revenue, the Global IPTV market is \$6.7 billion in 2009 and growing to \$19.9 billion in 2013, a compound annu-

al growth rate of 31%. By 2013, Europe and North America will generate a larger share of global revenue, due to very low ARPUs in China and India, the fastest growing (and ultimately, the biggest markets) in Asia. Figure 2 shows the global IPTV revenue forecast for 2009-2013

MRG's IPTV Global Forecast - 2009 to 2013 incorporates the most recent information on current IPTV deployments around the world, as well a forecast for IPTV subscribers, service revenue, and system revenue from 2009 to 2013. MRG breaks down the IPTV ecosystem into six markets: Access Systems, Video Headends, Video-on-Demand, Set-top Boxes, Middleware and Content Protection/Digital Rights Management (CP/DRM). In addition, we split up the market into 4 regions around the world: Europe, Asia, North America and Rest of World, as well as the Worldwide (global) market. contact Rob Smith at rsmith@mrgco.com, or visit www.mrgco.com



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IPTV News



IIF Delivers New IPTV Standards

November 17, 2009- The Alliance for Telecommunications Industry Solutions (ATIS) announced the release of three newly-created IPTV standards. ATIS' IPTV Interoperability Forum (IIF) - a global leader in IPTV standards development - recently completed this work.

These deliverables standardize multiple areas of the IPTV ecosystem. The three released standards consider: a test plan for validating objective quality models in the context of IPTV services; IPTV software

download sequence and remote management security considerations and requirements to ensure interoperability between service provider IPTV multicast applications; and network provider domains, home networks, and the IPTV terminal device.

The recently-released documents are detailed below:

- Test Plan for Evaluation of Quality Models for IPTV Services (ATIS-0800025) describes a comprehensive test plan for validating objective perceptual quality models in the context of IPTV services. The plan defines the procedure for evaluating quality models' criteria, performance, evaluation and documentation. Using a single standardized test for various services allows diverse algorithms to be consistently tested and enables easier cross-comparison. This document is intended to benefit Quality of Experience (QoE)- model standards development and alternatively evaluate QoE-model test processes.

- Remote Management of Devices in the Consumer Domain for IPTV Services (ATIS-0800009.v002) expands ATIS-0800009 to include additional details on security considerations and clarifies the software download sequence and protocols.

- Multicast Network Service Specification (ATIS-0800019) provides a baseline set of requirements to ensure interoperability between service provider IPTV multicast applications and the network provider domain, the home network, and the IPTV Terminal Function (ITF). It describes an IP multicast service that the network provider can use as a basis for a linear/broadcast TV service.

"These standards will enable interoperability between disparate elements of the vast IPTV ecosystem," said Susan Miller, ATIS President and CEO. "Successfully realizing remote device management and multicast network interoperability - while simultaneously ensuring high QoS levels across and between services - is essential to IPTV's continued growth and adoption."

These new standards are available through the ATIS Document Center at: <https://www.atis.org/docstore/>. The Document Center is an online resource for published and pre-published telecommunication standards, as well as technical reports, requirements and guidelines produced by ATIS-sponsored industry forums and committees.

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Endavo 3-Screen offers simpler, affordable and faster method of deploying video content globally

SAN JOSE, Calif. (Nov. 17, 2009) - Endavo Media, an Atlanta-based Internet TV Platform provider, showcases its 3-Screen content delivery solution this week at the Online Video Platform Summit, a featured event at Streaming Media West.

With this latest enhancement, producers, broadcasters and telcos using the Endavo Platform will be able to deliver their live and on-demand video to PC, mobile and television screens-all through a single hosted solution.

"People are turning in record numbers to computers, smart phones, and IPTV for video content," Endavo Media's CMO Peter Contardo said. "Endavo Media enables our clients to deliver their video everywhere-times three-with a 3-Screen reach at an affordable cost. It provides a faster method of

deploying content globally through Web sites, media players, live streaming and more-all to three screens."

The Endavo Media Internet TV Platform provides powerful ways to deliver live and online programming by including social media features, such as sharing of content, commenting, chatting, and much more, to connect with and grow the audience. This 3-Screen Platform also allows content producers to manage and distribute content, and multiple ways to monetize online video in order to generate new revenue.



Alpha Networks Presents Wireless HD Kit for IPTV/Connected TV Solution

December 10, 2009 - Alpha Networks announced that it will showcase its new Wireless HD Kit for IPTV/Connected TV Solution that features an optimal, cost-effective application for at-home distribution of multimedia content.

The new device can deliver four concurrent HD streams to any Wi-Fi enabled device, like IP-STBs, connected-TVs, or HD media players. This new solution enables IPTV service providers to cost effectively provide subscribers with HD service on multiple televisions. It also allows TV manufacturers themselves to offer consumers an exciting connected home and HD experience.

Based on 802.11 4Tx4R Implicit Beamforming MIMO technology, the wireless HD kit is able to perform over long distances for whole-home coverage with low latency and minimum packet loss. Equipped with a special real-time antenna diversity scheme, it effectively prevents channel fading and radio fluctuations. It also supports robust rate selection and TDM-like Scheduled MAC for guaranteed QoS without collisions. In addition, it supports UPnP/DLNA, easy installation, and is compatible with third-party client chipset solutions.

"Alpha Networks has the most comprehensive networking, wireless, and multimedia technologies and product portfolio in the networking ODM/OEM industry, and has long focused on digital home markets. This new wireless HD kit perfectly matches the emerging needs for HD video home networking. Alpha will bundle it with our digital home products, like HD media players and IP-STBs, to offer service providers and CE brand-name companies a complete home entertainment solution," said Mr. Tim Kang, Alpha's Chief Technical Officer.



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Espial Announces Newest Release of IPTV Middleware

Dec. 3 -Espial announced the newest release of the Evo TV Service Platform. Espial IPTV middleware is designed to serve millions of subscribers with a minimal capex investment, to provide a highly responsive user experience and to offer a powerful service creation environment. Field proven, highly scalable, based on open standards and supporting the interactive IP applications demanded by consumers - it is an ideal choice for IPTV and IP Cable operators.

The newest release of Evo TV Service Platform extends these capabilities with several major new features for telco and cable operators including:

- 3-screen support for new video-on-demand and subscription management services from PCs and mobile phones.

- Personalized UI advertising allows operators to build targeted advertising campaigns including static, clickable and video advertisements to drive new revenues and increase utilization of their value-added services.

- Time-Shift-TV services (TSTV) allow subscribers to take advantage of network-based Start-Over TV (Replay TV), Catch-up TV (Delayed or Look-back TV) and Pause Live TV.

- Arabic language and right-to-left character support.

- Professional-grade developer tools for rapid, easy and standards-based service creation using a PC-based integrated develop-

ment and preview environment.

- Operational enhancements to significantly reduce operational costs. Includes seamless upgrades, new CPE Management features, set-top box self registration and security enhancements reflecting an independent security audit.

"We're very pleased to launch this new version of Evo TV Service Platform. In response to market requirements, we're delivering these advanced features to better serve our IPTV and Cable IP customers," stated Robert Nadon, Director of Product Management. "With this release, we demonstrate our continued ability to meet the most demanding requirements of leading telco, cable and satellite TV service providers worldwide."



Ruckus Wireless Awarded Patent Essential to Running Video, Voice and Other Multimedia Applications Reliably Over Wi-Fi

December 7, 2009 - Ruckus Wireless announced that the United States Patent and Trademark Office (USPTO) and the European Patent Office (EPO) have each awarded the company a landmark patent on conversion of multicast to unicast transmissions. Multicast, a one-to-many transmission technique for data communications networks, is frequently used for delivering multimedia information to multiple simultaneous recipients. For example, IPTV service providers use it to deliver

IP-based broadcast television programming to subscribers over a broadband network; many enterprise applications such as tele-presence and video training also use it.

For 802.11 networks where multicast information delivery is not guaranteed, multicast-to-unicast conversion is essential to providing more reliable transmissions and more predictable performance for loss- and delay-sensitive multimedia applications.

The Ruckus multicast-to-unicast patent is particularly significant to the wireless networking industry as the growing demand for wireless multimedia communications coincides with the rise in Wi-Fi usage everywhere.

By converting multicast or broadcast packets into one or more unicast packets, as described in the Ruckus patent, Wi-Fi systems can now leverage the powerful 802.11 acknowledgment feedback mechanism. This built-in feedback helps to achieve extremely low packet loss rates for multicast traffic while increasing the number of multicast streams that the system can reliably transport.

When combined with intelligent antenna arrays, multicast-to-unicast conversion further enables the Wi-Fi systems to adapt automatically to changes in the signal environment. These systems can use unicast acknowledgements as one of many decision criteria to automatically "steer" Wi-Fi transmissions over the best performing signal paths in real-time.

Ruckus Wireless has commercialized Smart Wi-Fi solutions and remains the only Wi-Fi supplier with systems deployed in more than a million locations around the world, supporting the distribution of high-definition IP-based television over standard Wi-Fi.

"This patent fundamentally improves the quality of service that can be delivered over Wi-Fi networks," said Bill Kish, Ruckus Wireless CTO, co-founder and one of the named inventors. "New applications for Wi-Fi are quickly appearing as performance and reliability increase. There is incredible demand for wireless that is as reliable as a wire. Ruckus is taking it there."



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- Increase content consumption – by offering subscribers the ability to watch on their TV at home or on the go.
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- The WhereverTV web portal www.wherever.tv, where consumers can manage their content subscriptions, channel line-up and watch Internet video from within a community environment; and
- The WhereverTV widget, so that users may access their channel line-up at some of the most popular web applications, including iGoogle, Blogger, MySpace and Facebook.



For more information, contact: Chris Dion
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TV Advertising Series

Television Advertising

Television advertising is the sending of promotional messages or media content to one or more potential program viewers. The viewers are influenced by the messages, resulting in actions that benefit the advertiser. In 2010, many new TV advertising services will be offered by TV broadcasters, compared to the simple linear ad insertion practices used in traditional broadcast systems. Some may even be as simple as inserting ads into video on demand programming systems.

Advertisers benefit most by providing messages to people with an interest in their products or services. Media companies (such as broadcasters) are paid by broadcasters to send promotional messages. Advertisers (or their ad agencies) coordinate the selection of broadcasters and the transmission of promotional messages using advertising campaigns.

Broadcasters operate systems that gather, organize and provide people with the content they want to see. They may purchase license rights for the content through their systems, or create new (original) programming, and then merge it with advertising media. Viewer types, such as teens, adults, women and men, may be determined by the types of content being watched.

Viewers select the programs they want to view, most of which contain promotional messages that motivate them to take actions that satisfy advertiser's business objectives. Some such actions may include purchasing a product, signing up for an email campaign or subscribing to a service.

Figure 1.1 shows how the television advertising model works. The advertiser pays money to the TV broadcaster for the insertion of a promotional message. The TV broadcaster mixes in advertising messages with content that the viewer wants to watch. After the viewer watches the advertising message, they perform a desired action (buy a product or subscribe to a list), which earns revenue for the advertiser.

TV Advertising Marketplace

As the mix of media promotion types is shifting from broadcast television to Internet marketing, television viewing habits are changing,

This article is Part 1 of a 9 Part Series

Advanced TV Advertising List	Month
Television Advertising	Jan 10
TV Ad Types	Feb 10
TV Advertising Technologies	Mar 10
TV Advertising Systems	Apr 10
TV Ad Campaigns	May 10
TV Ad Production	Jun 10
TV Ad Metrics	Jul 10
TV Advertising Economics	Aug 10
TV Ad Regulations	Sep 10

and TV advertising web portals are changing how companies submit and manage TV ads.

Advertising Shift to Internet Marketing

Since 2005, advertising spending for TV broadcasting has not increased much while 30% of advertising budgets shifted to Internet marketing. It makes sense. It's easy, measureable and effective (targeted).

TV Viewing Habit Changes

Television viewing habits are changing. Viewers have developed ad blindness and widespread integration of digital video recorders (DVRs) has enabled them to bypass commercials. To overcome some of these challenges, some TV advertisers are migrating to paid placement advertising. Product placement within content prevents viewers from skipping past the ads, as they're embedded within the programs. While this is good for TV networks that produce and sell the programming, it can be bad for broadcasters because they don't make money on the paid placement advertising.

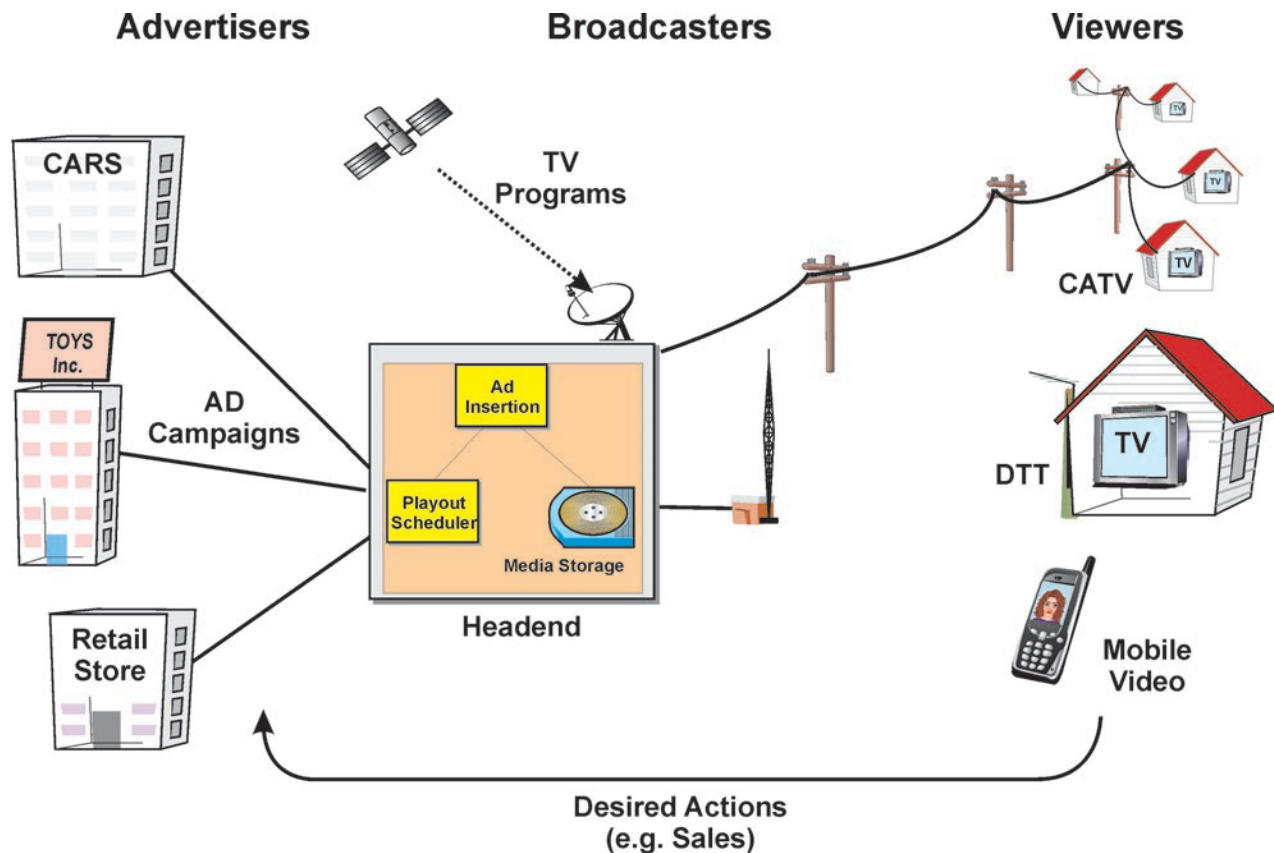


Figure 1.1, Television Advertising

TV Ad Bidding Web Portals

Television ad bidding web portals are web sites that allow companies or people to submit promotional media, such as television commercials and logos, and bid on the insertion of these ads into TV broadcast systems. In 2008, Google quietly entered into the TV advertising industry.

In addition to optimizing the ad revenue that can be generated, their web based advertising system streamlines the ad submission process. Some TV broadcasters can have content submission guidelines that are over 100 pages and many people at the broadcaster level are part of the approval process. TV broadcasters may require that ads be scheduled in ahead of time, sometimes weeks in advance. Web based TV advertising portals may allow advertisers to instantly adjust their ad campaigns (provided their ads are pre-approved).

Figure 1.2 shows a comparison between TV advertising and Internet advertising in 2009. This comparison shows that advertising

during the Superbowl in the United States costs approximately \$2.7 million per slot, which translates to 2 cents per viewer. This can be compared to pay per click Internet ads which earn approximately 54 cents per click.

TV Broadcasting Revenue Sources

Some key ways to earn revenue in TV broadcasting systems are to sell content, earn advertising revenue, or generate revenue from direct sales.

Content Revenue

TV broadcasters earn revenue from content through subscription fees and pay per view fees. In general, content costs from networks have been increasing. What is more challenging is that viewers can get content through many new media channels, which reduces the value of TV

➔ **Superbowl TV Ad is
\$2.7M = 2 cents per
impression**

➔ **Internet Ads Earn 54
cents per click!**

Figure 1.2, Television and Internet Comparison

networks. To keep and grow their viewership, TV broadcasters must get content that is more valuable to their viewers.

Advertising Revenue

Each year the amount of money that companies spend on advertising increases with the gross national product (GNP). Unfortunately for TV broadcasters, advertisers are shifting ad spending to the Internet. Today, approximately 1/3rd of the amount of TV ad spend (\$47B per year in the United States) is now being used for Internet marketing (\$17B per year). The growth in TV ad spending since 2009 has been approximately 0% [www.TVB.org], while the growth in Internet ad spending is well over 10% [www.IAB.org].

Television Commerce

Television commerce (T-Commerce) is revenue that is generated from direct sales which are processed through TV broadcasting systems. T-Commerce is shifting revenue from Internet systems (e-commerce), which had already exceeded \$1,000 per person per year in the United Kingdom, in 2007 [www.IAB.org]. Unfortunately, in 2010, many TV broadcast systems use proprietary systems that do not support T-Commerce and the costs and barriers to implement T-Commerce can be substantial, delaying its use in TV systems for several years.

Advertisers

An advertiser is a company or person that pays for services provided to others in return for the inclusion or presentation of marketing messages. Advertisers use marketing plans to define their promotional strategies, media communication objectives, and media channels (media mix). Advertisers define promotional projects (advertising campaigns), which determine the messages and audiences they want to communicate with (audience reach).

In general, advertisers want to motivate recipients to take action with the least promotional costs possible, so they test numerous promotional programs to measure both response and effectiveness.

Marketing Plan

A marketing plan contains the objectives of the marketing process, the responsibilities and incentives of those involved in the marketing process, and the resources that will be available or used for the marketing programs. A marketing plan may also define which media channels will be used (media mix).

Advertising Campaigns

Television advertising campaigns are marketing activities that send promotional messages to people about products, services and options that are offered by a company. TV advertising campaigns define how and when promotional message are provided to television viewers.

Audience Segments

Audience segmentation is the identification of viewers by categories (segments). TV programs may attract specific audience segment types.

TV Broadcasters

A broadcaster is a company that transmits or provides information to users that are connected to or able to access signals on the broadcast network. Broadcasters may provide a mix of linear (scheduled) programming, on demand programming and other services, such as gaming and communication applications. Some of the programming offered may include promotional messages (ad supported networks) and some may be provided without ads on a paid subscription basis.



Content Sources

The content for broadcasters may come from a mix of sources such as TV networks (affiliates), syndicates, content aggregators and original programming. Content often accounts for the highest cost within a TV broadcasting system.

Ad Supported Network

An ad supported networks is a provider of media programs that accept advertising time or ad spots as a form of compensation for the right to distribute (broadcast) content. Broadcasters may use a mix of direct advertising sales staff and advertising networks to obtain promotional advertising revenues.

Broadcast Systems

Broadcast systems can be a mix of transmission lines, processing equipment, and distribution systems that can acquire media from content sources and provide it to viewers. A TV broadcaster may use several types of distribution systems (such as cable, wireless broadcast,

satellite, or mobile telephone) to reach customers. Because it is unlikely that the same company owns all types of distribution systems, broadcasters are integrating their systems with one another.

Figure 1.3 shows a TV broadcasting model in which TV broadcasters link content providers (TV programs) to viewers through a variety of distribution system types. This diagram shows that a TV broadcaster receives content from a mix of sources. The headend selects and manages content that is sent through the distribution network. The distribution network transfers media to viewers and a TV broadcaster may transfer content through multiple types of distribution channels, such as satellite, broadcast, cable, mobile, data, and Internet. Viewers may access programming on multiple types of devices such as home televisions, mobile telephones, or multimedia computers.

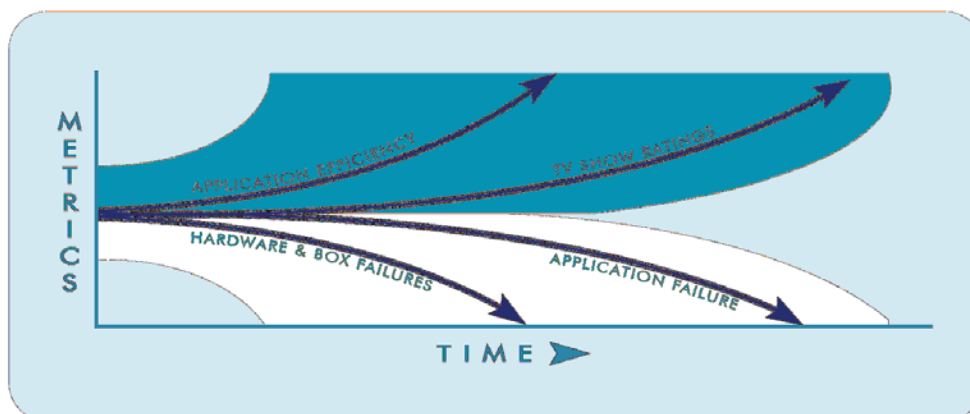
TV Viewers

Television viewers are people who watch or access viewable media products (such as television shows). TV viewers consume desired media based on personal preference. They are influenced by the media (content and ads) and perform actions that, without exposure to the media, they may not have otherwise performed.

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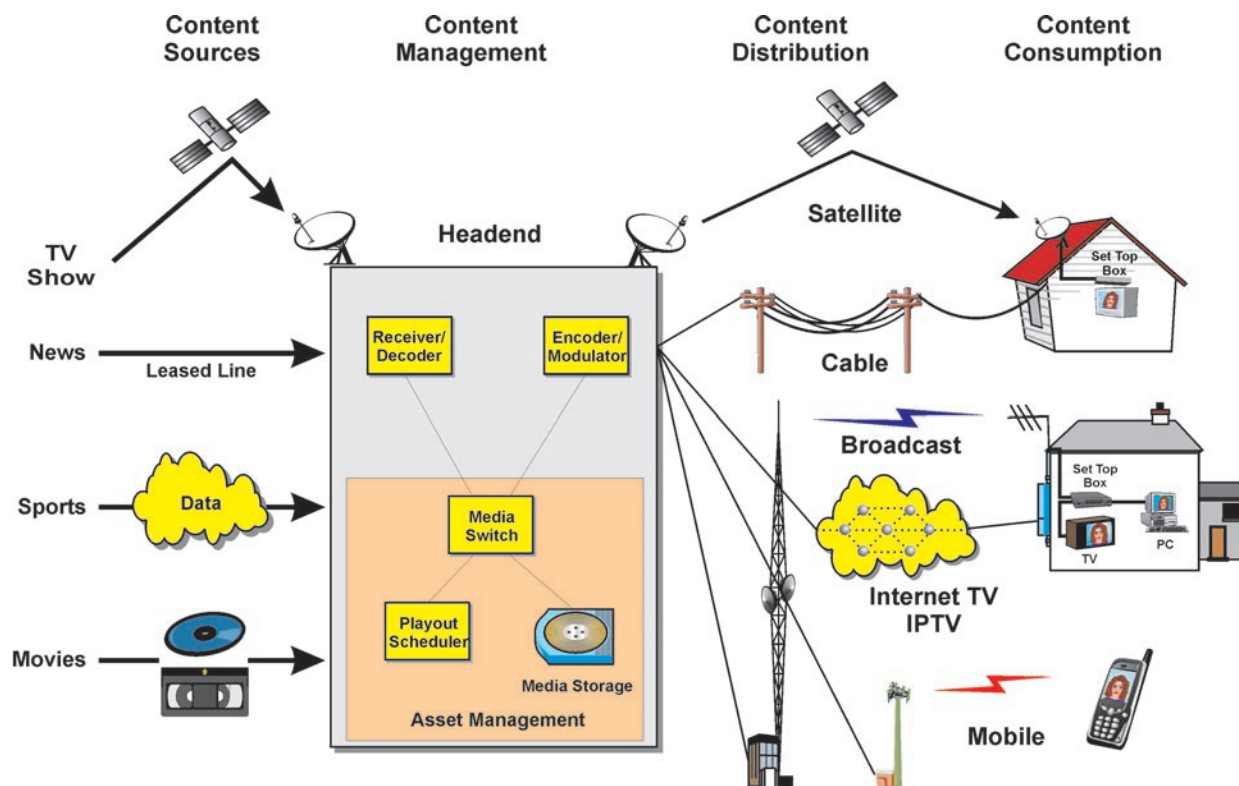


Figure 1.3, TV Broadcast Systems

Target Audiences

A target audience is a section of people who share common characteristics. These people may be more likely to be interested in buying or being associated with products that match audience needs or desires. By identifying desired target audiences, advertisers can pick which programs they desire to insert their advertising messages into.

Interruption Marketing

Interruption marketing is the unanticipated insertion of promotional messages into recipient media consumption activities (such as web browsing or television viewing). With the introduction of digital video recorders (such as TiVO), consumers may be able to store programs and skip through advertising messages that interrupt their programs.

Ad Blindness

Ad blindness is the ability of a viewer to ignore or not notice the presence of an advertising message. Even when the viewer does not skip past advertising messages, these ads may still be ignored by viewers who have become numb to their presence.



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Featured Article

IPTV Service Rates 2010 Update

By: Michele Chandler

As the demand for IPTV service has increased, IPTV service providers have adapted by offering an increased number of channel bundles and diverse product offerings. Since we have been tracking IPTV service rate plans in August 2006 and October 2008, we have noticed that while demand has increased, in the past year, IPTV rate plan costs have remained steady.

Customizable Service Plans

IPTV service providers have made rate plans more customizable for customers at the same rates. Now TV, now offers 28 free channels to subscribers, up from 15 free channels in 2008, with the option of adding channel bundles that include sports, movies, music, travel and more. Subscribers can also purchase individual channels ranging from news to specialized TV channels such as STAR Cricket. Surewest Communications, an IPTV service provider in California, United States, has also introduced an A La Carte option for channel bundles such as cinema in addition to other services such as DVR, TiVo, and On Demand programming. As per previous research, adult TV channels are still the most expensive channel bundles to purchase.

Figure 1 shows sample IPTV service rate plans. IPTV service rate plans are commonly offered in basic, mid-level and premium rate

plans. Some packages have special offers or can be ordered with other services such as channel bundles.

New Service Types

New services have been included with service plans such as online data storage in addition to previous content offerings including TV gaming, DVR, caller ID, and voice bundles. Aon TV subscribers have access to 300 radio stations and can play personal photos, videos and music files directly on their TVs.

More HDTV Channels

HDTV channels have been added to IPTV service rate plans since 2008. Customers have the option of buying service plans with a pre-determined number of HD channels, or buying a channel bundle with additional HDTV channels, movies, and music videos.



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Company	Basic	Mid-Level	Premium	Extras
AT&T U-Verse <i>United States</i>	U-basic \$19.00/mo Up to 20 channels	U200 \$64.00/mo Up to 230 Channels	U450 \$109.00/mo Up to 390 channels	-Most bundles, excluding U-basic, waive \$95 installation fee. -Most come with Total Home DVR, excluding U100 and U-basic -Only U450 comes with HD service for free, otherwise it's an extra \$10/month
Verizon FIOS TV <i>United States</i>	FiOS TV Essentials Over 250 Channels, Up to 14 HD \$47.99/month	—	FiOS TV Extreme HD Over 320 Channels, Up to 65 HD \$57.99/month	-First month free with Essentials and Extreme HD -Activation at no cost -Home Media DVR free for 3 months
SureWest Communication <i>California, United States</i>	Digital Basic 77 Channels \$35.99/Month promotional rate (\$52.99/Mo. standard rate)	Digital Choice 169 Channels \$45.99/Month promotional rate (\$67.99/ Mo. standard rate)	Digital Encore Total Channels: 178 \$48.99/Month promotional rate (\$73.99/ Mo. standard rate)	-SureWest offers plans for HDTV, On Demand, and has A La Carte options
Aon TV <i>Austria</i>	aonTV Basic Package 70 TV Channels € 4.90/ month	—	aonTV Premium An additional 24 channels including aonTV Basic € 7.90/ month	-TV packages include integrated Video Library with over 300 movies on demand -aonTV video subscription available for € 9.90/ Month - aonTV HD Video Library available for € 4.90/ Month
SwissComm <i>Switzerland</i>	SwissComm TV Basic More than 140 Channels CHF 19/month	—	SwissComm TV Plus More than 140 Channels, plus live sports, Over 130 radio stations, Live pause TV channels and films in HD CHF 29/month	-SwissComm TV Plus offers Swisscom TV-Guide, Minor Protection and the option of multiple devices

Figure 1: IPTV Service Rate Plan Comparison

International Programming Choices

International service plans are still popular amongst IPTV service providers. Providers such as Verizon continue to offer La Conexión for their Spanish-speaking customers. To meet the demands for international content, companies such as SingTel, which offers mio TV, has service plans for entertainment in English, Chinese and Indian consumers. SuperSaver Chinese, one of the mio TV plans, offers television programming from Korea, Taiwan, Hong Kong, Japan and Singapore.

Equipment Rentals

Fees for set top boxes, installation, equipment rental and monthly membership service are still prominent amongst service rates. In many

cases, companies are willing to waive these fees as a special promotion in return for signing up for a service plan. Start up fees often require a fixed line network connection and Internet access with the service provider.

Many IPTV service providers have introduced the option of purchasing or renting more than one set top box for their household, increasing accessibility to their services. SwissComm offers customers the option of multiple devices with their premium package Swiscomm TV Plus.

Since costs and number of channels still vary by location, it is still difficult to compare IPTV service rate plans since they are based upon location. Since October 2008, several IPTV service rate plans have offered the same services for the same prices, while some have introduced more channels for their subscribers.

Featured Article

Internet Set Top Boxes

By: Mike Galli, MRG, Inc. Analyst

Internet TV Set-top Boxes allow viewers to watch streaming media from the Internet on standard television sets. Internet TV STBs (often called I-STBs) may be dedicated Internet TV STBs or they may provide functions that are added to other media processing devices such as gaming consoles, Blu-ray players or even TV sets. (Note: Internet TV is also known as Internet Video or Over-the-Top (OTT) service.)

I-STBs may be designed or programmed to work with Internet TV Broadcasters or they may allow the users to directly connect to the media source (such as an Internet streaming channel directly from a TV Broadcaster).

I-STBs can have a mix of network connection options, output connection types, operating systems, media processing types, and pro-

ocols. The choice of specific I-STBs can determine which service can be provided to the viewer and how efficient the system can operate.

One of the main things to keep in mind is that I-STBs need to be as inexpensive as possible. Back when algorithms for encoding and decoding video were being looked at, it was decided that the algorithm needed to be asymmetrical meaning that the encoders should be more complex and costly because there are much fewer of them and decoders needed to be simple and inexpensive as there are a lot of them. So MPEG, or more precisely the Discrete Cosine Transform (DCT), was chosen for these reasons. Other algorithms such as Wavelets (JPEG 2000) are more symmetrical and are not a good choice for consumer video applications.

ISTB Connections (Multiple OTT Devices)

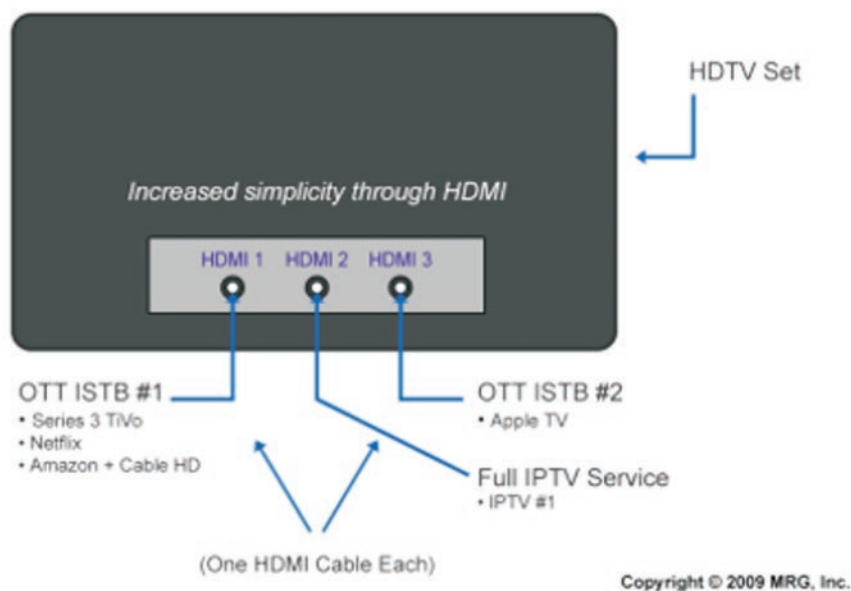


Figure 1: HDTV: Rear Panel Connections Simple OTT/IPTV Installation



Some I-STBs are fairly closed systems such as Roku and TiVo, so these are fairly simple to control from a development and cost stand point. Some of the newer ones such as the Boxee box and the I-STB from Triveni Multimedia are more open and have more challenges in both of these areas. The advantage for the more open I-STBs is that the customer is not confined to a "walled garden." The consumer can do quite a bit more.

Network Connections

I-STBs commonly have a wired Ethernet (RJ-45) and/or Wi-Fi for the Broadband connection. Some I-STBs combine other types of network connections such as DTT Broadcast TV, Cable TV (RF inputs) or Satellite (also an RF input).

Wired Ethernet connections are fairly simple these days and have

very little impact on the STB itself. Wi-Fi wireless components are a bit more complex, especially those used for streaming video. One of the more popular chip manufacturers is Atheros. Its products offer a number of the 802.11 extensions, as well as QoS. These become necessary if you have 2 or more video streams at the same time or if you have other applications such as VoIP in addition to normal Internet activity.

The combination or hybrid STBs include interfaces that allow you to add a traditional service along with an Internet service. This can be combined into one service such as with the TiVo Series 2 or 3 products so there is one interface and you don't have to switch back and forth on your TV set inputs.

Figure 2 shows a comparison of I-STBs, IPTV STBs and Cable STBs. When I-STBs first came out they were not nearly as capable as

	Network Interface(s)	Streaming Protocol(s)	File Format(s)	Codec(s)	DRM, CA, Encryption
I-STBs					
TiVo ¹	100 MB Ethernet, B/G Wireless	MPEG-2 PS	MPEG-2 "long GOP" CBR, VBR	MPEG-2	
Roku	100 MB Ethernet, B/G/N Wireless	MPEG-2, ASF	MPEG-2 TS, WM, VBR and CBR Support	MPEG-2, WMV3, VC-1	WMDRM 10, Playready DRM
Netgem N8000	100 MB Ethernet, DVB-T or DVB-C (QAM), B/G/N Wireless	HTTP, RTP/RTSP, MPEG2 TS, Progressive Download, P2P	MPEG-2/4 CBR, MPEG-2/4 VBR	MPEG-1, MPEG-2, MPEG-4 Part 2 & 10, WM, VC-1	Nagra, Verimatrix, WMDRM 10, AES Encryption
Apple TV ²	100 MB Ethernet, B/G/N Wireless	Quicktime (RTP/RTSP), Download	Quicktime	MPEG-4 Part 10	Fairplay DRM (iTunes)
Cable STBs					
Motorola DCT 2000	QAM 64/256	MPEG-2 TS	MPEG-2 CBR, VBR	MPEG-1 and 2	DigiCipher 2, DVB-CA
IPTV STBs					
Amino 110	100 MB Ethernet	MPEG-2 TS	MPEG-2 CBR, VBR	MPEG-1 and 2	SecureMedia, Verimatrix, Irdeto, others

Figure 2: Internet STBs Compared with IPTV and Cable STBs



Figure 3: Rear View of the Roku HD STB Source: Roku

they are today. Many of them come close to conventional STBs. In the case of Netgem's I-STB, it is actually the same product with certain features enabled or disabled.

Output Connections

The outputs of I-STBs typically include composite video (RCA), stereo audio (RCA), and S-Video. Some also include component video, · HDMI, and optical audio. These can be PAL or NTSC depending on the country.

Figure 3 shows the rear connections on the Roku HD STB. (Roku also offers an SD only model that is less expensive.)

Some models have a USB connection which can be used for external storage or for an external Wi-Fi adaptor.

Operating System

The I-STB operating system and its available drivers coordinate the overall operation of the device. Some of the available operating systems for I-STBs include Linux, Windows, and Android.

Interface applications (clients) are installed into the STBs to provide a user interface (navigation and programming guide), middleware access (media mapping), access control (conditional access), and content protection (copy protection or encryption).

Media Decompression

One of the main functions of the I-STB is to decode the streaming media (video, audio, images and other data). The media decompression may be performed by special chips (codecs) or it may be performed by software programs that are used by a general microprocessor. When a STB can perform processing in a dedicated codec chip, the decoding process may be faster and reduce the burden of processing on the system microprocessor, but the range of decoding options is reduced to whatever the chip supports. As time goes by and defacto standards emerge it is expected that all STBs will use dedicated chips such as those from Broadcom, ST Micro or NEP.

There are several types of media compression that may be used in Internet TV systems. In general, the higher the compression ratio (lower bandwidth required), the more processing is required by the STB. Some of the common types of compressed video include MPEG-2, MPEG-4, Flash and Windows Media Video (VC-1). Some of the common types of compressed audio include MP2, MP3 and AAC.

MPEG has different profiles and levels that determine the decoding and presentation abilities, but only a few are used for Internet Video. The main differences are with the choice of resolutions and aspect ratios. Right now the main drive is to be able to deliver a resolution of 1080p with a 16:9 aspect ratio so people with HDTV sets will not be disappointed.



Hybrid IPTV STB	Full Broadcast Service + Managed IP (IPTV Video) (e.g. BT, Verizon)	Used most in Europe, North America and Asia.
Standalone OTT ISTB	Internet Interface to IP (unmanaged) (e.g. Apple TV, Roku, VUDU)	Gaining popularity in North America and Europe.
Hybrid OTT ISTB	Full Broadcast Service + Internet Connection to IP (unmanaged) (e.g. Netgem)	Starting to appear as an alternative to hybrid IPTV.

Source: Copyright © 2010 MRG, Inc.

Figure 4: IPTV vs. OTT STBs

Internet Video/OTT is produced either with a single stream or with multiple streams. In the case where there is a single stream, the job of the STB is very simple. In cases where there are multiple streams, the STB communicates with the server to determine which stream, based on how much bandwidth is available, to decode. Most of the intelligence is at the server. There is a client on the STB that monitors the buffer and looks for underflow or overflow. This information is fed back to the server which then decides which of the stream rates the network can handle.

The lowest data rates are typically around 700 Kbps which are perfectly fine for an SD TV set and possibly for a small HDTV, but for larger HDTV sets the necessary data rates can be as high as 4 Mbps. In this case, the consumer must have a very high-speed Broadband service.

Processing Capability

When developing an I-STB two of the main considerations are processing power and memory. If it is using a software decoder then the majority of the processor is used for decoding video and this can vary quite a bit from the previous section. This is another reason why dedicated chips are preferred by STB manufacturers.

To give you a better idea here are some scenarios:

SD video at 700Kbps could be achieved on an x86 with something like a 500MHz Celeron processor. It may be possible to squeeze as much as 25% more performance with fine tuning of the software decoder to the CPU and other network and system software customizations.

An ARM processor will require almost 1GHz and a MIPS processor around 800MHz.

By comparison an I-STB with a dedicated video decoder chip would be able to get by with as little as 200MHz ARM CPU.

HD video at 2-4 Mbps would require a 2GHz CPUx86 at least with some support from a graphics card.

Protocols

STBs use protocols to send and process commands and to provide vital information to other components, primarily the streaming server.

Protocols usually have multiple versions and later versions commonly add capabilities.

Control

Control protocols are used to setup and manage the operation of the STB device. The most common protocol used for video streaming is RTSP. RTSP is used to set up the video session and provides for "trick-play" functions. Trick-play functions are; Pause, Rewind and Fast-forward. An SNMP agent might be included for network maintenance and control, but this is uncommon.

For Internet connection management these are primarily in the Ethernet and Wi-Fi components and don't impact the other portions of the I-STB, but are important for establishing and maintaining network connectivity.

The multicast control protocol (IGMP) is used by IPTV STBs as the Broadcast streams are multicast on a managed video network, but are not used yet in Internet applications. First, in the Internet space everything is assumed to be on-demand, for now at least. This will change as more and more "live" streams are deployed. Most ISPs block multicast traffic so this will require the cooperation of them to enable this.

Internet STBs do use a CA/DRM client to authenticate the user and to decide what content the user is entitled to.

Storage

Memory considerations are looked at in two areas: RAM/Flash memory and whether or not to include a hard drive.

RAM/Flash Memory

RAM/Flash memory depends on the number of clients needed (see protocol section) and what is needed to assist in decoding. RAM memory can be a big help with buffer overflow especially in cases where there isn't a hard drive in the STB. The operating system is typically held in RAM memory.

Disk Memory-Hard Drives

Disk memory (such as a hard disk drive) may be included to store programs and other media (such as downloaded ads), but are commonly used for downloading video and providing PVR functions.

Remote Control Devices

Just like other video devices IR remote controls are very common. Blue Tooth (commonly used for wireless mice and keyboards) tends to be a limiting choice as the average U.S. living room is big enough to be out of range of these devices. If the STB is in a different room, then RF is used. For the fancy gyro-based products, like those from

Logitech and Hillcrest, RF is used as they cannot guarantee line-of-site.

I-STBs have come a long way and are getting better and better. Consumers will certainly have a lot to choose from.

Conclusion

The I-STB is part of the rise of OTT (Internet Video) that MRG started tracking in 1999 with the growth of CDNs. As an 11-year old market sector, OTT, with the assistance of standalone and embedded I-STBs, will continue to offer consumers, content owners and service providers ever new opportunities to access and distribute high quality and high value content to TV sets around the world.

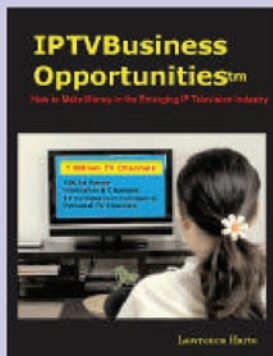
For more information on MRG's OTT and Hybrid STB reports, see:

<http://www.mrgco.com/iptv/ott09.html>

<http://www.mrgco.com/iptv/hstb09.html>



Mike Galli, MRG Analyst, has over 20 years experience in the video industry. He has worked in engineering, marketing and sales positions at The Grass Valley Group, VXtreme (acquired by Microsoft), DIVA, Minerva and Kasenna (acquired by Espial). Mike has a BSEE degree from the University of California, Berkeley.



IPTV Business Opportunities™

by: **Lawrence Harte**

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IPTV Channel Hosting

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EGIHOSTING.com was founded to serve the unique needs of high bandwidth customers. Their network was engineered specifically to provide reliable and affordable bandwidth to

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830 Hillview Ct, Suite 255, Milpitas,
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<http://www.egihosting.com>

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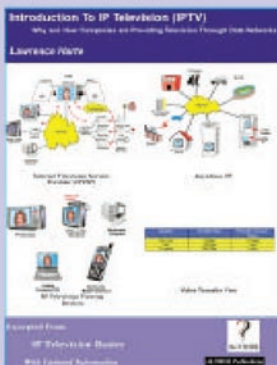
GravityLab Multimedia offers hosting, encoding, delivering live and archived on demand streaming media content. GravityLab has expert level understanding and experience with building, deploying and managing content delivery networks for audio and video media hosting.

Box 3016, Eugene, OR 97403,
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www.gravlab.com

GRID-TV

Grid-TV is a developer and host for Internet TV networks. Grid grants international broadcasting licenses, enables the usage of international broadcasting networks and sells broadcast planning software. The company operates the International Playout Center, the German Internet TV broadcasting centre, from where up to 10,000 Internet TV stations can be controlled at the same time. FileLoadBalancing neuronal net, TV-Edit and TV-Serve technologies developed by Grid-TV enable unique, target-group specific, global TV based on common internet streaming software over all communication channels such as satellite, terrestrial, cable, UMTS, GPRS and internet.

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www.grid-tv.com



Introduction to IP Television (IPTV)

by: Lawrence Harte

This book explains how and why people and companies are using IP television services and how global television services are already available through broadband data networks. You will learn how IP Television works and the different types of viewing devices that can be used in IPTV systems such as standard televisions with adapters, dedicated IP televisions, multimedia computers and mobile telephones.

ISBN: 1-932813-35-7

Price: \$14.99 print version or \$12.99 eBook !!!

www.AthosBooks.com/iptv.html



InterLake Media

InterLake System provides managed server hosting, complex IT integration and business process outsourcing with a focus on server operations and integration of third party systems. InterLake Media provides global media streaming, video player platforms and technical integration for online video projects to a customer base of international media and advertising companies.

Försterweg 2
14482 Potsdam
Germany
+49 89 38665360
www.interlake.net

JayJay TV

JayJay TV is a high speed web TV platform for musicians, producers and anyone involved in the creative industry. Music videos, artist interviews, live performances and TV shows will be available to watch on their player. JayJay TV also offers a high end video production service with access to studios, models and makeup artists.

www.jayjaytv.com

NeuLion

NeuLion develops and provide programs for Internet Protocol Television. The NeuLion iPTV Platform enables the building of streaming video applications, encoding and delivery of video and broadcast television. NeuLion specializes in speeding to market the content provided by our partners and lowering their costs of delivery by using the Internet.

1600 Old Country Rd,
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www.neulion.com

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NextBiT Computing is a leading provider of Intellectual Property, product solutions and services catering to hardware design, operating systems, media frameworks, multimedia system software, content management, streaming, broadcasting and rights management softwares for Multimedia Consumer Electronic, broadcast, and handheld devices.

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SofTV

SoftTV is an application that allows you to watch TV on your computer for free. You will be able to view streaming broadcast television shows as well as listen to radio stations from around the world over your internet connected computer. The channels are provided by the participating broadcast stations and then streamed to your player.

www.softtelevision.com

SOFTing Ltd.

SOFTing d.o.o. is an International company offering a wide range of specialized services in the fields of Streaming media, IPTV, System integration and Custom software development. They are a European pioneer in the Internet TV market, being one of the first companies in Europe to launch an highly innovative Internet TV software platform. SOFTing strives for transparency in its business through shared management, the quality of its products, its ethics and security, with focus on respect for the Human Being. Through these actions it reflects all the simplicity of not only a great but also a happy and productive company.

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www.eu-softing.com

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www.streamhoster.com

Streaming Media Hosting

Streaming Media Hosting provides content delivery services for thousands of businesses and consumers worldwide. Streaming Media Hosting's proprietary technologies empower customers with advanced capabilities to manage, deliver and monitor audio, video, and multimedia content over the Internet. Streaming Media Hosting's services for small businesses are unique because they integrate the Industry's most popular formats: Adobe Flash, Microsoft's Windows Media, Apple's QuickTime Media, RealNetworks' RealMedia, MP3 and Java Streaming, with Streaming Media Hosting's proprietary technologies and robust online management tools, to provide customers with a level of simplicity, and features unparalleled in the Industry. Streaming Media Hosting excels above all others by providing the complete streaming solution.

177 Riverside Ave. Suite 241,
Newport Beach, CA 92663,
United States
Tel: +1-800-963-4347
www.streamingmediahosting.com

TV1.DE

TV1.DE is a leading European IP Television Service provider. Headquartered in Munich, Germany, TV1 delivers Advanced Internet Television services for Broadcast, Enterprise, Retail, and Government organizations worldwide.

Beta-Str. 9a
Unterfohring, D-85774
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Fax:+49-89-960-570-79

TVU Networks

TVU networks is a global live TV service that enables TV Broadcasters and private individuals to broadcast TV channels to a global audiences over the Internet. TVU uses a new application-level multicasting technology (similar to peer-to-peer file sharing) that allows broadcast costs to be exponentially lower than those of today's streaming technology.

Asionics Hi-Tech Center,
1279 Zhong Shan Rd west,
Shanghai, P.R., 200051
www.tvunetworks.com

TV Worldwide

TV Worldwide is a leading global Internet broadcasting and streaming media company, TV Worldwide is developing a network of video channels that is an affiliation of community-based Internet television stations, each underwritten by a strategic partner, "aimcasting"(SM) to targeted demographic audiences worldwide. TVWorldwide.com works with strategic partners to develop the latest in live and archived state-of-the art video streaming content applications.

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Fax:+1-703-961-9255
www.TVWorldwide.com

ViaStreaming

ViaStreaming offers live audio and video media streaming services. Their plans are made up of different combination of bandwidth, formats (Shoutcast, aacPlus, Windows Media Audio and Video) and listener options to suit your budget and target audience.

www.viastreaming.com

VBrick

VBrick is a leader in Enterprise IP Video solutions. VBrick solutions work over standard IP networks and the Internet to deliver rich media communications that connect people everywhere -- from employees and customers, to partners and shareholders. Our comprehensive product suite and end-to-end solutions are used in a wide range of live and on-demand applications including meeting and event broadcasts, distance learning, digital signage, TV distribution, video surveillance, and Web-based marketing campaigns.

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www.videopros.com

Vividas

Vividas is a developer and provider of video technologies which enable full screen, high quality live or pre-recorded video to be delivered via the internet or over corporate networks from standard web servers. There is generally no requirement for a first time user to install player software. Their proprietary technology overcomes the disadvantages of competing solutions that typically offer only partial screen or poor quality full screen viewing and generally require the user either to have or to install specialist player software.

For corporations, we deliver a high quality live stream to desktops across the Wide Area Network using existing web server infrastructure without any adverse effect on network performance.

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Fax:+44-0-20-7936-9100
www.vividas.com

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www.convergenceindia.org

New Products

Home Media Networks

Atheros Communications AR7400

The AR7400 chipset is the world's first solution designed to comply to the IEEE 1901 draft 2.0 standard (now elevated to Sponsor Letter Ballot status), a global, open powerline standard. The Atheros Powerline chipset consists of the AR7400 MAC/PHY transceiver and the AR1500 analog front end (AFE) / line driver. The chipset also supports operation up to 75 MHz, utilizing all of the practically available spectrum defined in the 100 MHz IEEE 1901 charter, and avoiding the global, high-power FM radio broadcast bands from 76 to 108MHz. It has an integrated ARM11 core processor and includes MII, RGMII, and GMII to support 10/100/1000 Ethernet PHYs along with UART and SPI serial interfaces for Smart Grid applications. It supports both SDRAM and DDR memory and includes a high-performance, hardware-based packet classification engine to support carrier-grade quality of service (QoS), dynamic channel adaptation and channel estimation to maximize throughput in harsh powerline conditions.

Mailing Address: 5480 Great America Parkway, Santa Clara, CA 95054, United States
Tel: +1-408-773-5200 Fax: +1-408-773-9940
www.atheros.com

Home Media Networks

Ralink Technology RT3883 and RT3593

RT3883 and RT3593 are the industry's first commercially available single-chip 450Mbps 3x3 802.11n access point and station solutions with Beam Forming technology. The RT3883, featuring three dual band 2.4/5GHz radios, a 500MHz MIPS74K CPU, and a variety of connectivity interfaces, is the most advanced Wi-Fi Access Point/Router platform in the industry. The RT3593 is the most highly integrated 450Mbps 3x3 client solution available on the market with an embedded media access controller, baseband processor, and three dual band 2.4/5GHz radios on a single chip. Both RT3883 and RT3593 chips incorporate Ralink's high-performance radio and baseband architectures, delivering best-in-class Wi-Fi performance, low power consumption, and cost-effective price points. The RT3883 and RT3593 chips are also the first single chip 802.11n solutions which support the IEEE standard's optional Beam Forming functionality.

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www.ralinktech.com

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Media Players

ViewSonic VMP80

The ViewSonic VMP80 media player, powered by the Yahoo! Widget Engine. The VMP80 will enable existing HDTV owners to view movies, TV shows, web videos, photos, go shopping, play games and more with TV Widgets. ViewSonic's VMP80 media player may be inserted in-line between the TV and the set top box, eliminating the need to toggle between the Internet and cable or satellite box. Via an existing IP connection, Internet content and services can be overlaid with broadcast programming from the video provider to a ViewSonic or any HDTV. By simply pressing a button on the remote control, users can bring up the TV Widget Dock, browse through content and select the service they wish to enjoy without missing a moment of their favorite TV program.

Mailing Address 381 Brea Canyon Road, Walnut, CA 91789-0708, United States
Tel: +1-909-444-8888 Fax: +1-909-468-1240
www.viewsonic.com

Set Top Boxes

MIPS Technologies Android™ 'Home'

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Mailing Address: 955 East Arques Avenue, Sunnyvale, CA 94085, United States
Phone: +1-408-530-5000 Fax: +1-408-530-5150
www.mips.com.

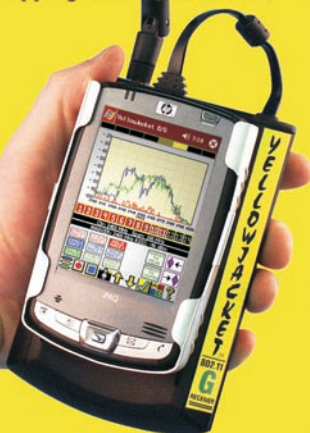
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NTCA Wireless Symposium 2010	1/13/2010	1/15/2010	Tampa, United States	www.ntca.org
Internet Telephony Expo East 2010	1/20/2010	1/22/2010	Miami, United States	www.tmcnet.com/voip/conference/
OPASTCO Winter Convention 2010	1/23/2010	1/27/2010	San Diego, United States	www.opastco.org/site/meetings/winter/
ACUTA Winter Seminar 2010	1/24/2010	1/27/2010	New Orleans, United States	www.acuta.org/home.cfm
NAPTE 2010	1/25/2010	1/27/2010	Las Vegas, United States	www.natpe.org/natpe/
Blackhat DC 2010	1/31/2010	2/3/2010	Arlington, United States	www.blackhat.com/html/bh-sponsors/sponsors.html
CeBIT 2010	3/2/2010	3/6/2010	Hannover, Germany	www.cebit.de/cgc_e
SCTE Canadian Summit 2010	3/9/2010	3/10/2010	Toronto, Canada	www.scte.org
Media Summit 2010	3/10/2010	3/11/2010	New York, United States	www.digitalhollywood.com/MediaSummit.html
International Wireless Communication Expo 2010	3/10/2010	3/12/2010	Las Vegas, United States	www.iwceexpo.com/iwce2010
Media Summit (2010)	3/10/2010	3/11/2010	New York, United States	www.digitalhollywood.com/MediaSummit.html
Carriers World Asia 2010	3/17/2010	3/19/2010	Hong Kong, Hong Kong	www.terrapinn.com/2010/cwa/
CTIA Wireless 2010	3/23/2010	3/25/2010	Las Vegas, United States	www.ctiawireless.com/
IPTV World Forum 2010	3/23/2010	3/25/2010	London, United Kingdom	www.iptv-forum.com/
MiPTV 2010	4/12/2010	4/16/2010	Cannes, France	www.milia.com
NAB 2010	4/12/2010	4/15/2010	Las Vegas, United States	www.nabshow.com
InterOP 2010	4/25/2010	4/30/2010	Las Vegas, United States	www.interop.com/lasvegas/
Digital Hollywood Spring 2010	5/3/2010	5/6/2010	Santa Monica, United States	www.digitalhollywood.com
Streaming Media East 2010	5/11/2010	5/12/2010	New York, United States	www.streamingmedia.com/east
ISCE 2010	6/7/2010	6/10/2010	Braunschweig, Germany	www.isce2010.org/
Broadcast Asia 2010	6/15/2010	6/18/2010	Singapore, Singapore	www.broadcast-asia.com
Cinema Expo International 2010	6/21/2010	6/24/2010	Amsterdam, Netherlands	www.showest.com/filmexpo/cinemaexpo/index.jsp
Blackhat Las Vegas 2010	7/24/2010	7/29/2010	Las Vegas, United States	www.blackhat.com/html/events.html

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Month Deadline/ Submission	Special Editorial Focus	Featured Articles	Buyers Guide	Trade Show Participation
January 10 12/29/09 1/5/10	Internet TV STB	Internet TV Station Viewer Ad Profiles	IPTV Channel Hosts	CES Future TV Show NTCA Blackhat
February 10 1/26/10 2/4/10	TV Widgets	TV Widget Consortiums Screening UGC	TV Widget Developers	Digital Switchover Strategies
March 10 2/26/10 3/3/10	Shared Middleware	IPTV System Sharing	IPTV Middleware	IPTV World Forum Carriers World
April 10 3/25/10 4/2/10	Internet TV Broadcasting	Adding Internet TV to IPTV Systems	Internet TV Systems	NAB MiPTV Interop
May 10 4/24/10 5/1/10	Home Media Networking	Home Media Network Standards	IP STBs	NCTA Digital Hollywood
June 10 5/23/10 5/30/10	IPTV Customer Care	Automating Customer Care	IPTV Customer Care Systems	Broadcast Asia
July 10 6/23/10 6/30/10	Virtual Headends	Virtual Headend Options	Virtual Headend Systems	Blackhat SIGGRAPH
August 10 7/21/10 7/28/10	TV Content Licensing	TV Content Licensing Strategy	Content Licensing Agents	
September 10 8/18/10 8/25/10	TV Program Guides	TV Program Guide Data	TV Metadata Management Systems	IBC
October 10 9/19/10 9/26/10	Multiplatform Advertising	Adapting TV Ad Delivery	Ad Delivery Systems	Supercomm NECA Interop Broadcast India
November 10 10/18/10 10/25/10	TV Ad Monitoring	TV Advertising Metrics	TV Ad Validation Systems	Interop Telco TV
December 10 11/18/10 11/25/10	Broadband TV Quality	Managing Quality on Internet TV	Internet TV Testing Equipment	

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